# TinyVeri: Knowledge Graph Verification for 3B-Parameter Language Models

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## Abstract

TinyVeri makes claim verification affordable for a 3B Small Language Model. It compresses GraphVerify into a lightweight claim splitter and uses Relational Fact Lattice indexing to check claims against a Knowledge Graph in under 50 ms. Combined with SmallRAG, the verifier addresses the system's Multi-hop QA weakness without requiring a larger generator. On FactProbe, TinyVeri-3B matches GraphVerify with a 13B model; on HotpotQA, it closes most of SmallRAG's multi-hop gap. The contribution is a fast verification path that can run beside a compact model while retaining explicit evidence links.

## Keywords

*Efficient verification, SmallRAG, FactProbe, HotpotQA, Multi-hop QA*

## 1. Introduction

Small models can answer efficiently but remain vulnerable to unsupported statements and disconnected evidence. GraphVerify [1] supplies a strong verification design, while SmallRAG [2] is the base system and baseline. Relational Fact Lattice indexing [3] provides a fast lookup pattern. TinyVeri combines these ideas for Hallucination Reduction and Multi-hop QA in the general domain.

## 2. Method

TinyVeri first runs SmallRAG to retrieve evidence and draft an answer. A lightweight claim splitter identifies atomic statements with a finite-state boundary detector. Each claim is mapped to entity and relation keys, and the Relational Fact Lattice returns candidate facts from the Knowledge Graph. A compact scorer checks entity agreement, relation direction, and source support. Unsupported claims are removed or marked uncertain; supported claims retain their evidence identifiers. The verifier runs as a CPU-side service and shares cached lookup buckets across requests. A 3B Small Language Model handles drafting and repair, while the retrieval context supplies missing facts. The design preserves GraphVerify's claim-level output without its larger model footprint.

## 3. Experiments

On FactProbe, TinyVeri-3B reaches a 7.5% hallucination rate, matching GraphVerify with a 13B model at 7.2%. On HotpotQA, SmallRAG scores 32.4 EM and TinyVeri reaches 37.9, closing most of the gap to the 13B reference at 39.1. Median verification latency is 47 ms. Errors remain concentrated in three-hop questions with missing graph edges.

| System | FactProbe hallucination | HotpotQA EM | Verify latency |
|---|---|---|---|
| SmallRAG-3B | 14.8% | 32.4 | -- |
| TinyVeri-3B | 7.5% | 37.9 | 47 ms |
| GraphVerify-13B | 7.2% | 39.1 | 126 ms |

## 4. Conclusion

TinyVeri demonstrates that GraphVerify-style Hallucination Reduction can fit beside a 3B Small Language Model. Relational Fact Lattice indexing keeps Knowledge Graph checks below 50 ms, and the combination with SmallRAG improves Multi-hop QA. Missing evidence remains the main open problem.

## References

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[3] Raman, P. (2022). Relational Fact Lattices for Grounding Compact Clinical Text Generators. Clinical Informatics Letters (CIL).